PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 10 · Cognitive biases in sports

TUESDAY OCT 27 AND THURSDAY OCT 29, 2026

GET OUT A SHEET OF SCRAP PAPER.

Write your name at the top. You'll answer a few questions on it today, and I'll collect it at the end of class.

UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your personal repository sits next to spax402-course-materials, the shared folder Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

> Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials. ↵

3 Confirm that today's deck is in spax402-course-materials/decks/.

The Case Study and BRIEF.md arrive through the same /update.

? WHAT'S THE PROBABILITY THAT MILWAUKEE SWEEPS THE WORLD SERIES?

01

FAST AND SLOW THINKING



THE NEXT QUESTION IS A COMPETITION TO SEE WHO CAN ANSWER FIRST.



A BAT AND A BALL COST $1.10.
THE BAT COSTS $1.00 MORE THAN THE BALL.
HOW MUCH DOES THE BALL COST?

STRENGTH: HUMANS RECOGNIZE PATTERNS.

Weakness: Humans recognize patterns when there are none.

SYSTEM 1 VS SYSTEM 2 THINKING

SYSTEM 1

Intuitive, emotional, and fast.

Most reliable in “Kind Environments” in which feedback is reliable, immediate, and repeated. Best applied when decisions are time sensitive or the stakes are low.

SYSTEM 2

Analytical, deliberate, and slow.

Most useful in “Wicked Environments” in which feedback is noisy, infrequent, or delayed. Best applied when decisions are complex, high-stakes, and data-rich.

Source: Daniel Kahneman, Thinking, Fast and Slow (2011)

SYSTEM 1 VS SYSTEM 2 THINKING

ENVIRONMENT	APPROPRIATE SYSTEM
A batter predicts the pitcher's pitch type from the pitcher's windup and release.	
A GM predicts player work ethic based on an interview.	
A strength coach determines which lifts should be prioritized to mitigate injury risk.	
A coach decides whether to bench a player after multiple poor shooting performances in a row.	
A coach decides whether to go for it on fourth down.	

ISRAELI ARMY INTERVIEWS

1. In the 1950s, Israeli army candidates went through field test to predict how well a recruit would do in the army.
2. The interviewers conducted 15-20 minute interviews and then assigned a score.
3. Follow-up analysis revealed these scores had almost no correlation with the recruit's success.
4. Kahneman was asked to develop a better system to evaluate officer candidates that would not take any additional time.

Source: Daniel Kahneman, Thinking, Fast and Slow (2011)



HOW WOULD YOU FIX IT WITHOUT ADDING ANY TIME?

02

JUDGING PEOPLE

A linebacker prospect, described by his position coach:

Intelligent, hard-working, impulsive, critical, stubborn, envious.

? **ON A SCALE OF 1 TO 10, HOW MUCH WOULD YOU WANT HIM IN YOUR LOCKER ROOM?**

REAL ESTATE "EXPERTS"

1. Real estate agents were shown identical photos, description, neighborhood data for a listing.

2. Some were asked to estimate the house's value without seeing a list price. They had a mean evaluation of $120k.

3. Others were shown one the following list prices: $119,900 | $129,900 | $139,900 | $149,900

Northcraft, G. B., & Neale, M. A. (1987). Experts, amateurs, and real estate: An anchoring-and-adjustment perspective on property pricing decisions.

Average estimate with no list price: $120k

Low list price: $119,900

High list price: $149,900

? **WHAT WAS THE AVERAGE EVALUATION FOR THOSE WHO SAW THE HIGH AND LOW LIST PRICE?**

AWARENESS OF COGNITIVE BIASES ISN'T ENOUGH.

We need systems to combat them.

ANCHORING BIAS

The tendency to weight the first piece of information you receive too heavily.

EXAMPLES IN SPORT

1. Evaluating a point-guard's decision making skill.
2. Reevaluating a player who was drafted highly.
3. The first number offered in a negotiation.
4. Pre-season conceptions of a team.
5. Success of a play early in a game.
6. Success of a player early in the season.

Another linebacker prospect, described by his position coach:

Envious, stubborn, critical, impulsive, hard-working, intelligent.

? **ON A SCALE OF 1 TO 10, HOW MUCH WOULD YOU WANT HIM IN YOUR LOCKER ROOM?**

HALO EFFECT

The tendency to let one positive trait or outcome create an overall favorable impression in other traits.

EXAMPLES IN SPORT

1. Assuming a good student in the classroom will have a high IQ on the field.
2. Thinking that a player who is good with the media is a good locker room leader.
3. Thinking that a coach who runs an intelligent scheme must be making the right decision on fourth down.
4. Projecting a point guard with a strong voice to be a confident in-game decision maker.

SHOULD A 5’10″ QUARTERBACK BE DRAFTED LOWER THAN A 6’3″ QUARTERBACK WITH THE SAME COLLEGE PRODUCTION?

STEREOTYPE BIAS

When beliefs about a group of people shape perception of an individual.

EXAMPLES IN SPORT

1. Player attribute evaluations being influenced by someone's ethnicity, nationality, build, handedness, etc.

2. Base rates built from who got a chance.

   Few quarterbacks under six feet were drafted high or handed the starting job, so "short quarterbacks don't succeed" rests on a small sample the stereotype itself filtered.

3. The sample changes when teams take the chance.

   Drew Brees and Russell Wilson were drafted lower partly on height. Kyler Murray (2019) and Bryce Young (2023) went first overall at about 5'10".

03

JUDGING DECISIONS

Super Bowl XLIX, February 1, 2015.

Seattle trails New England 28–24, second and goal at the 1, 26 seconds left, 1 timeout. Russell Wilson throws a slant and Malcolm Butler intercepts it.

? **WAS IT A BAD CALL?**



I'LL GIVE YOU $1,000 IF A COIN FLIPS ON HEADS. YOU GIVE ME $1 IF IT'S TAILS. YOU TAKE THE OFFER. WE FLIP IT AND IT'S TAILS. DID YOU MAKE THE WRONG DECISION?

OUTCOME BIAS

Judging a decision based on its result rather than the quality of the decision itself.

EXAMPLES IN SPORT

1. Coach retention after an unlucky season relative to run differential.
2. A coach reduces aggressiveness after a failed two-point conversion due to a dropped pass.
3. A play-action pass when the run-game is working is considered great play calling when it results in an explosive play, “over thinking it” when it results in an interception.

HINDSIGHT BIAS

Thinking an event was predictable after it happened.

EXAMPLES IN SPORT

1. Coaches feel “we already knew that” when reviewing data in a scouting report.
2. Scouts have overconfidence in their ability to evaluate talent.
3. Usually, fans, media, and owners are too critical of decisions that go poorly, not appreciative of decisions that go well.
4. Occasionally, fans, media, and owners give too much praise to poor risky decisions that turn out well.
5. “Once you know the outcome, you can’t reconstruct what you knew before you knew it.”

   Thinking, Fast & Slow

AFTER THE 2018 SEASON

16
8 ▼



Matthew Stafford
Detroit Lions · QB

2022 PRESEASON

Rank
39
—



Baker Mayfield
Carolina Panthers · Year 5

2025 PRESEASON

Rank
31



Daniel Jones
Indianapolis Colts · Year 7

AFTER THE 2018 SEASON

16
8 ▼



Matthew Stafford
Detroit Lions · QB

2021 PRESEASON

Rank
5



Matthew Stafford
Los Angeles Rams · Year 13

2022 PRESEASON

Rank
39
—



Baker Mayfield
Carolina Panthers · Year 5

2025 SEASON

ESPN

Perception of Mayfield shifting amid MVP-type run

Asked Wednesday about the MVP talk and whether he has paid much attention to it, Mayfield said he tries "not to get too high or too low" and he...



2025 PRESEASON

Rank
31



Daniel Jones
Indianapolis Colts · Year 7

2025 SEASON

Horseshoe Heroes

Colts' Daniel Jones is redefining expectations in the NFL MVP race



The Boston Globe

Why does playoff pressure get to some pitchers but not others?

It's an age-old question. We saw it on display once again with Madison Bumgarner in his brilliant 3-0 shutout over the Mets in the National...

Oct 8, 2016



MLB.com

Price shuts down doubters with stellar Series

LOS ANGELES -- David Price had heard the criticisms, the questions, the doubts about his ability to perform on the big stage.

Oct 29, 2018



ATTRIBUTION BIAS

Overemphasizing personal traits such as skill or character, while under emphasizing situational factors or luck (unless it's your mistake).

EXAMPLES IN SPORT

1. Individual player evaluations overly tied to team wins & losses.
2. A GM attributing their hits to skill and their misses to bad luck.
3. A coach attributing a converted 2-point attempt to instinct or playcalling, and a failed attempt to the nature of the game.

04

STREAKS AND CHANCE



IS THE HOT HAND REAL?

? FLIP A FAIR COIN FOUR TIMES. LOOK ONLY AT THE FLIPS THAT COME RIGHT AFTER A HEADS. ON AVERAGE, WHAT SHARE OF THOSE FLIPS ARE HEADS?

MILLER AND SANJURJO (2018)

FIVE HEADS IN A ROW



What is the chance the next flip is heads?

Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)

FIVE HEADS IN A ROW



Still 50%. The coin has no memory of the last five.

Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)

THE 1985 TEST

Treat each shot like a flip: H is a make, T is a miss.



Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)

THE 1985 TEST

Treat each shot like a flip: H is a make, T is a miss.



Made 3 of the 6 shots ○ right after a make: **50%**

1985 compared this number to 50%. Close to 50% meant no hot hand.

Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)

EVERY WAY FOUR FAIR FLIPS CAN LAND



Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

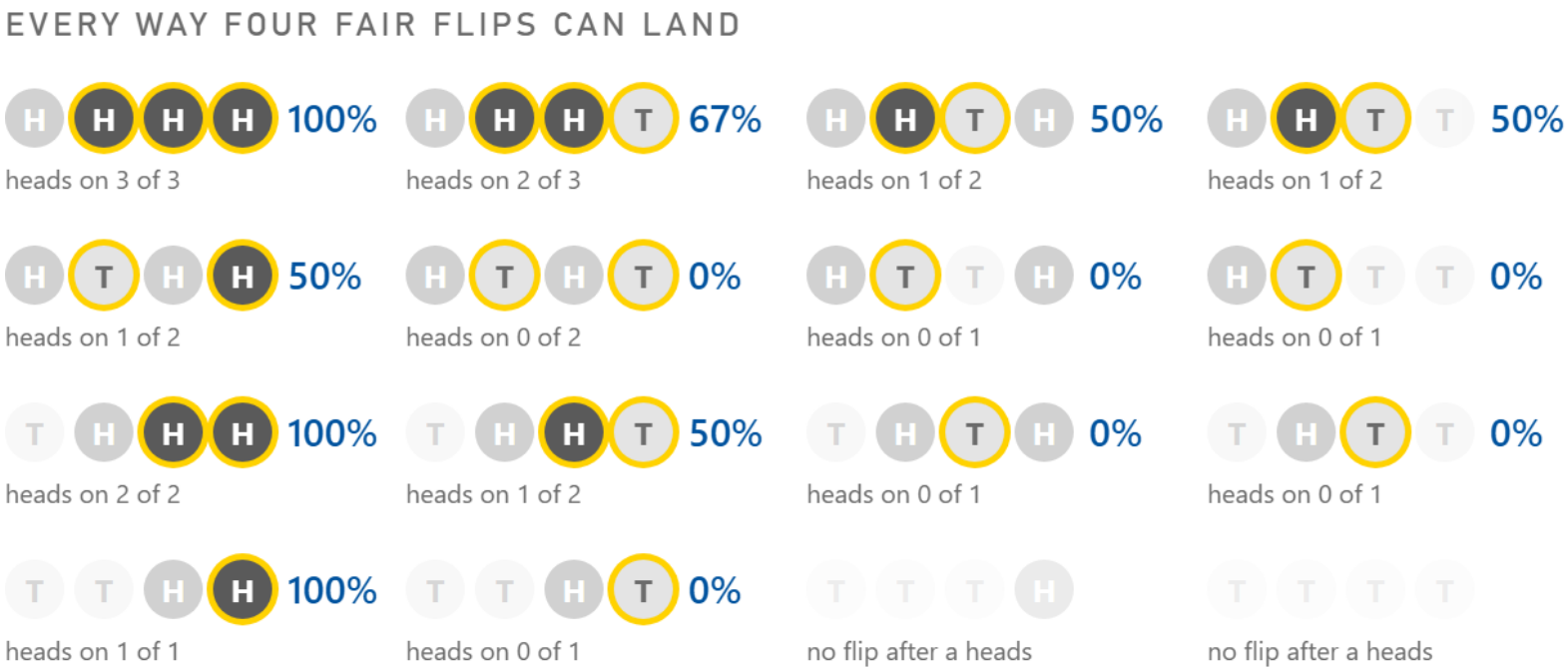
MILLER AND SANJURJO (2018)

EVERY WAY FOUR FAIR FLIPS CAN LAND



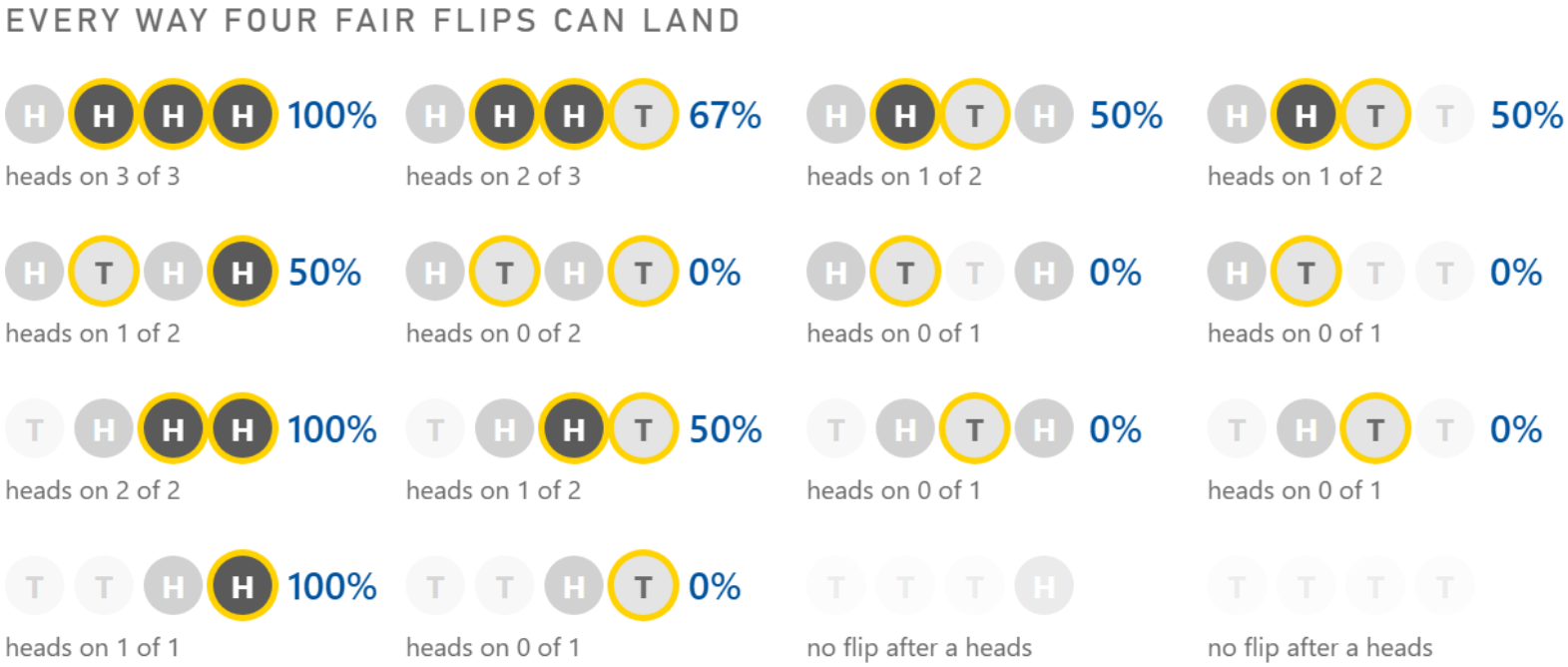
Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)



Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)



Average the 14 **blue** scores, the 1985 number for a coin:

40.5%

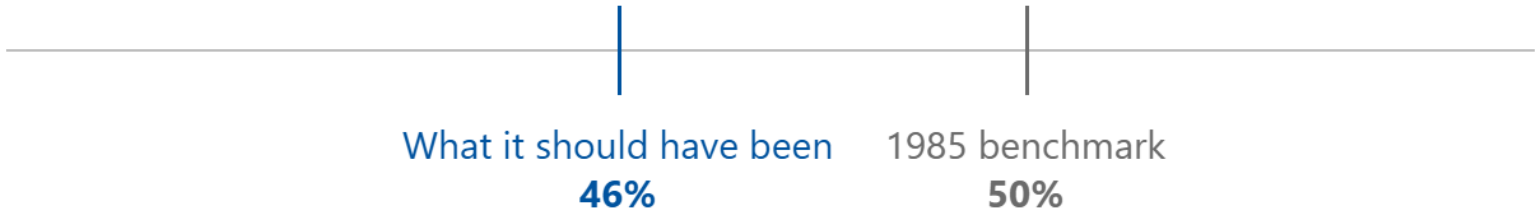
Every flip is still 50/50: 12 of the 24 ○ flips are heads. The 1985 score should sit below 50% if there was no hot hand.

Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

MILLER AND SANJURJO (2018)

THE RIGHT BENCHMARK

A 50% shooter with no hot hand, 100 shots: make % right after 3 straight makes



Shooting your usual 50% after three makes beats the 46% a shooter with no hot hand would get.

The 1985 study compared shooters to 50% when the right benchmark was 46%, so it missed a real hot hand of a few percentage points.

Source: Miller, J. B., & Sanjurjo, A. (2018). Surprised by the Hot Hand Fallacy? A Truth in the Law of Small Numbers. Econometrica, 86(6), 2019–2047.

RECENCY BIAS

The tendency to weight the most recent piece of information you receive too heavily.

EXAMPLES IN SPORT

1. A baseball player is determined to be “locked in” after a hot week.
2. A basketball player is said to have a “hot hand” after hitting a few three-pointers in a row.
3. Overweighting a teams end of year performance heading into the playoffs.

? A CAREER .300 HITTER IS 0 FOR HIS LAST 20. IN HIS NEXT AT BAT, IS HE MORE LIKELY THAN USUAL TO GET A HIT, LESS LIKELY, OR THE SAME?

GAMBLER'S FALLACY

The mistaken belief that past events are negatively correlated with future events.

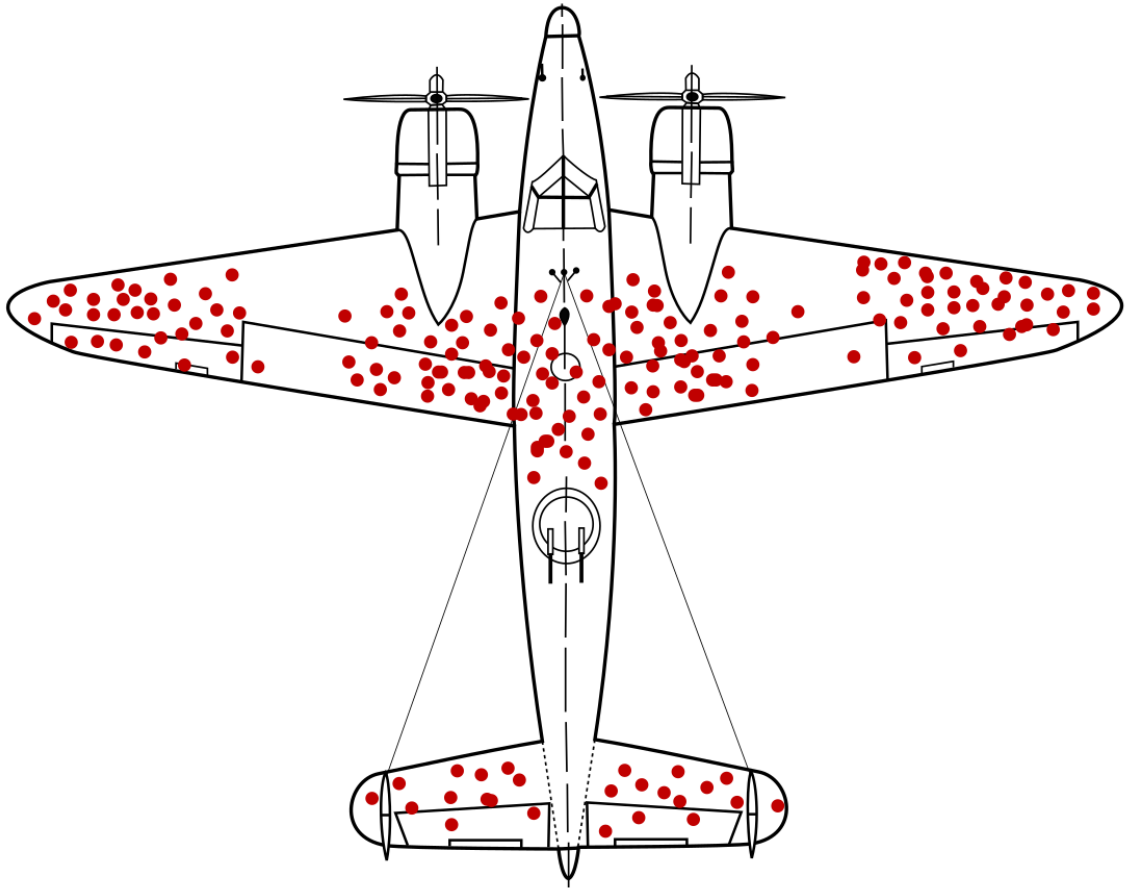
EXAMPLES IN SPORT

1. When a player makes ten free throws in a row, we believe they are “due” for a miss.
2. This mirrors the hot hand fallacy, but both stem from human discomfort with chance and tendency to create stories to explain chance.
3. Some stats are negatively correlated with others. We have a human tendency to believe even independent or positively correlated events are negatively correlated.

05

SURVIVORSHIP

WHERE SHOULD ARMOUR BE ADDED?



Source: Wikimedia Commons (2016). Survivorship Bias Diagram.

ARE SHORTER PEOPLE NATURALLY BETTER SHOOTERS?



Source: The New York Times (Branch, 2016). “Stephen Curry Sees Room for Improvement. Seriously.”

SURVIVORSHIP BIAS

The error of drawing conclusions from a group that has "survived" a selection process, while ignoring those who did not.

EXAMPLES IN SPORT

1. A strength coach directs young athletes to mimic elite professionals' training routines, not realizing those routines work for survivors who had rare genetic durability.
2. A study of what allows long-term veterans to achieve a long-career excludes professional athletes with short-careers, that may have burnt out.
3. Taking career advice from successful GM's who tell you "invest your 20s into your work, it'll be worth it.", while failing to interview anyone who pursued front office positions and failed.

88.5%

of NBA GMs picked Kobe Bryant when asked which player they would want taking a shot with the game on the line.

Carmelo Anthony, Tim Duncan and Steve Nash tied for second at 3.8% each.

Source: NBA GM Survey.



WAS KOBE CLUTCH?

06

STICKING WITH IT

? I HAVE A RULE THAT YOU HAVE TO GUESS.

"2, 4, 6" FOLLOWS MY RULE.

GIVE ME ANY THREE NUMBERS AND I'LL TELL YOU WHETHER THEY FOLLOW MY RULE.

CONFIRMATION BIAS

The tendency to search for, interpret, and remember information in a way that confirms your existing beliefs, while ignoring or discounting evidence that contradicts them.

EXAMPLES IN SPORT

1. Focusing on plays that reinforce your initial player evaluation.
2. Searching for data that supports, rather than informs, your opinion.

Last year, you signed a running back for 3 years at $10M a year, $5M a year guaranteed.

He badly underperformed in year one.

A free agent running back costs $4M a year, and you're confident he will outplay your starter going forward.



SHOULD YOU CUT YOUR STARTER AND SIGN THE FREE AGENT?

SUNK COST FALLACY

Continuing a course of action because of past investments,
rather than current evidence of future costs and benefits.

EXAMPLES IN SPORT

1. Starting your highest paid goalie instead of your backup who you believe is better.
2. Sticking with a player who has a large buyout, even when cutting your losses would be the optimal decision.
3. NBA: Higher draft picks got more minutes and longer careers than their play earned.

   Staw & Hoang (1995), Administrative Science Quarterly
4. NFL, the other side: the Broncos cut Russell Wilson in 2024 and took a record $85 million in dead money.
5. It is beneficial for owners to create environments where their front offices are comfortable owning up to their mistakes.



A COIN FLIP. HEADS, YOU WIN $200, TAILS YOU LOSE $100. WOULD YOU TAKE IT?

HOW OFTEN DO NFL TEAMS GO FOR IT ON FOURTH AND 1 NOW?

David Romer, Journal of Political Economy, 2006:

NFL teams should go for it on fourth down far more often than they do.

? **IF THE MATH SAID GO FOR IT IN 2006, WHY DID COACHES WAIT MORE THAN A DECADE?**

LOSS AVERSION & STATUS QUO BIAS

Losses loom larger than equal gains, so we prefer the current state of affairs even when a change has the better expected outcome.

EXAMPLES IN SPORT

1. Football: Conservative Fourth Down Decisions
2. Baseball: Saving your closer for a situation which may not come.
3. NBA: Refusing to use a challenge for a first-half three-point play.
4. Across Sports: Choosing a “safe” prospect over a high-upside one.
5. Fan Sentiment: Concern over rule changes. (e.g. Pitch Clock).

? A COACH SAYS: “WHEN I PRAISE A PLAYER AFTER HIS BEST GAME, HE’S WORSE THE NEXT GAME. WHEN I CHEW HIM OUT AFTER HIS WORST, HE’S BETTER.”

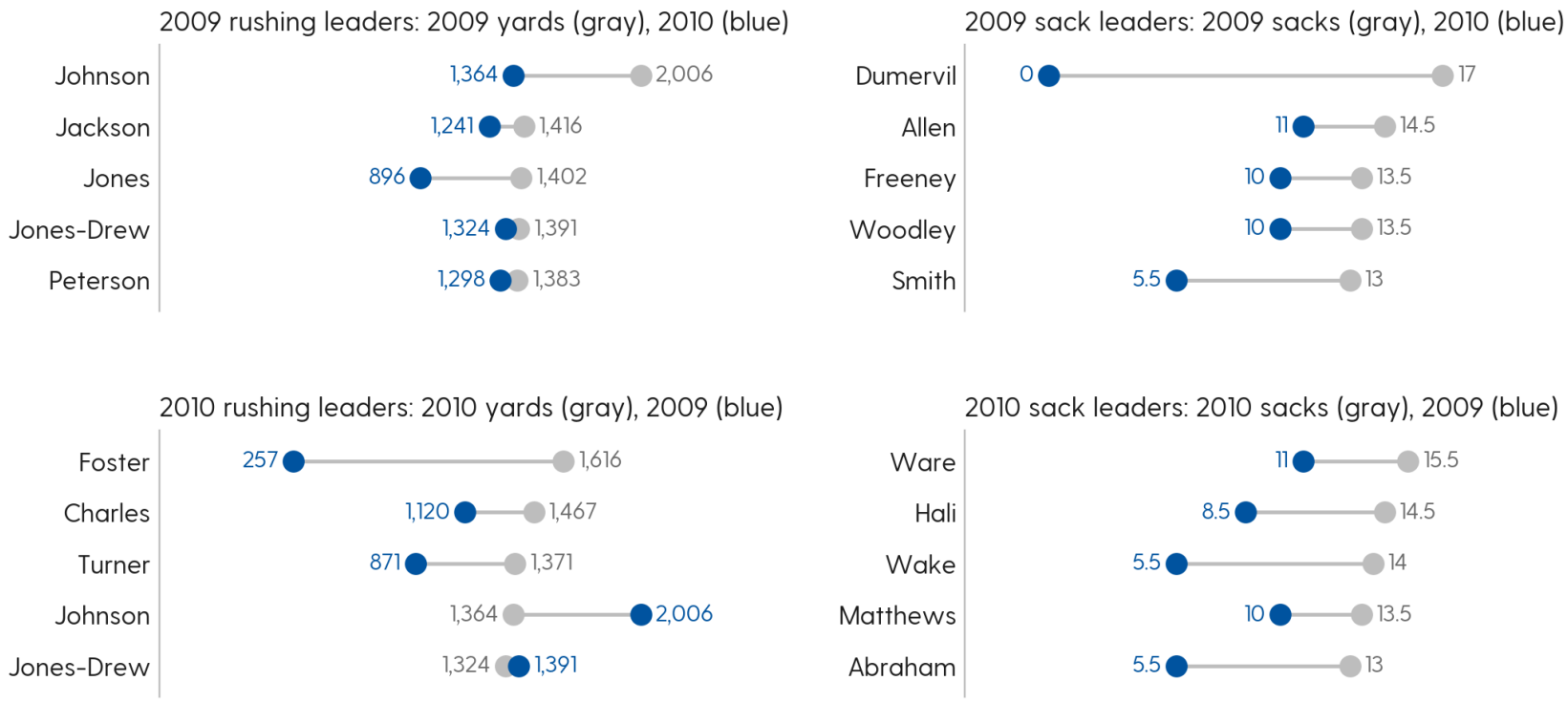
IS HE RIGHT THAT YELLING WORKS BETTER THAN PRAISE?

07

REGRESSION TO THE MEAN

? FT% AND 3P% ARE POSITIVELY CORRELATED. IF A PLAYER RANKS 2ND IN FT%, WOULD WE EXPECT THEM TO BE RANKED HIGHER OR LOWER IN 3P%?

REGRESSION TO THE MEAN



Year over year statistics are just related variables, like FT% and 3P%. We expect regression to the mean.

Severini (2020), Analytic Methods in Sports

REGRESSION FALLACY

Creating a story to explain the natural return of extreme performances toward average to a specific cause.

EXAMPLES IN SPORT

1. When a rookie has a positive outlier in year #1, there is a belief that they regressed in year #2.

2. Coaches thinking that berating players improves performance, while praise results in decline.

3. Teams that benefit from turnover luck one season are overrated going into the next season.

4. Some stats are more “sticky” than others. They should be analyzed individually to determine how predictive they are of future results.

08

POTENTIAL BIASES

POTENTIAL BIASES

ENVIRONMENT	POTENTIAL BIASES
A GM predicts player work ethic based on an interview.	
A strength coach determines which lifts should be prioritized to mitigate injury risk.	
A coach decides whether to bench a player after multiple poor shooting performances in a row.	
A coach decides whether to go for it on fourth down.	



ARE ANALYSTS LESS BIASED THAN SCOUTS?

09

THIS WEEK'S WORK



WHAT PROBABILITY DID YOU GIVE MILWAUKEE TO SWEEP?

CASE STUDY

A BIAS IN THE WILD

STEPS

1. Find a decision or article from the 2025 or 2026 season in which at least one cognitive bias was demonstrated.
2. Have Claude find public data that tests it, and build one interactive chart that makes your argument.
3. Commit the code, the chart and a link to the decision or article.
4. Answer the brief in half a page to a page.

BRIEF QUESTIONS

1. Which cognitive bias does your decision or article show, and what in it convinced you? Could a different bias explain it just as well?
2. What is the strongest argument for the decision as it was made, and does your data answer it?
3. What other data would you want before telling the decision maker they got it wrong?

Written to the coach, general manager or writer behind the decision.

FINAL PROJECT PROPOSAL

1. **Research question.**
   What you want to know. Should be one sentence.

2. **Potential strategies for answering the question.**
   The model or analysis you'd start with.

3. **Identified data sources.**
   Where the data comes from and how you'll get it.

4. **Expected challenges.**
   What could stop you or limit the answer.

FINAL PROJECT PROPOSAL: AN EXAMPLE

1 Research Question

What is the “True” 3-point ability for every DI NCAA Men’s Basketball Player?

2 Potential Strategies for Answering the Question

Create a linear regression model which predicts a player’s 3-Point percentage for the second half of a season based on their historic shooting efficiency. Include separate independent variables for more recent data to see if it should be more heavily weighted. Adjust for lurking variables such as their team’s offensive efficiency, shot volume, and shot difficulty.

3 Identified Data Sources

Stathead, Synergy, Kenpom

4 Expected Challenges

Stathead.com does not make data extraction easy. Small sample sizes result in high levels of variance. Inability to make predictions for players who are early in their career.

BEFORE YOU PUSH ON MONDAY, NOVEMBER 2

Commit your Case Study code, chart and link.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, November 2 at 11:59pm.